

THE SAMENESS LATTICE OF A CATEGORY: WEAK-EQUIVALENCE STRUCTURES, GALOIS FUNCTORIALITY, AND COMPARISON INTERVALS

RYOTA SHIBUSAWA

ABSTRACT. To each small category C we associate its *sameness lattice* $\text{Same}(C)$: the complete lattice of all weak-equivalence structures on C , that is, classes of morphisms containing the identities, closed under composition, and satisfying two-out-of-three. We develop Same as a categorical invariant. It is a complete lattice (Proposition 2.1); for a group G , viewed as a one-object category, it is exactly the subgroup lattice, $\text{Same}(G) = \text{Sub}(G)$, and for a groupoid the lattice of wide subgroupoids (Theorems 3.1, 3.3), so that non-distributivity already appears at the Klein four-group and the realization problem for Same contains the subgroup-lattice representation problem. It decomposes over connected components as a product of lattices, does not determine C within a component (the free composition triangle and V_4 share the lattice M_3), and is in general non-modular (A_4); on groups its distributivity is Ore’s local-cyclicity condition (Theorems 4.1–4.3). The assignment is functorial: a functor $F: C \rightarrow D$ induces a Galois connection $F_! \dashv F^*$ of sameness lattices, and $(GF)^* = F^*G^*$, $(GF)_! = G_!F_!$, so that $\text{Same}: \mathbf{Cat} \rightarrow \mathbf{CLat}_{\text{Gal}}$ lands in Galois connections (Theorem 5.2). A comparison of two structures on one category factors through the interval they span (recording strictly more than the comparison of their localizations), and a functor factors through the localization at its largest inverted sameness $V_F = F^*(\text{Iso}D)$; this factorization need not be conservative in general, but is conservative whenever V_F admits a calculus of fractions, and is an equivalence exactly when F is that localization (Theorem 6.1, Propositions 6.2–6.3, Theorem 6.4, Proposition 6.5). We read the settled part of the cubical-site phase diagram — which cube sites present spaces — as a map into $\text{Same}(\text{Presh}(S))$ (§7), and we sketch an object-oriented reading in which the Structure Identity Principle is one level (§8). All structural claims are proved; all finite claims are verified by exhaustive computation.

2020 *Mathematics Subject Classification*. 18A99 (primary), 06B15, 18N40, 55U35 (secondary).

Key words and phrases. localization, weak equivalence, two-out-of-three, complete lattice, subgroup lattice, Galois connection, comparison of theories, cubical site, Structure Identity Principle, behavioural equivalence.

Corresponding author. E-mail: r-shibusawa@daiichi-koudai.ac.jp.

1. INTRODUCTION

A recurring situation is that one object admits two descriptions, or two theories admit a translation, and one asks *whether they present the same thing*. In each case “the same” means a choice of which morphisms count as identifications: a class of weak equivalences, a localization. This paper takes *the collection of all such choices* as the primary object and studies it as a functorial invariant of a category.

Definition 1.1. A *sameness* (or *weak-equivalence structure*) on a small category C is a class $W \subseteq \text{Mor}(C)$ such that

- (i) W contains all identities;
- (ii) W is closed under composition;
- (iii) (*two-out-of-three*) for composable f, g , if two of $\{f, g, g \circ f\}$ lie in W then so does the third.

$\text{Same}(C)$ is the set of samenesses ordered by inclusion; $W_1 \subseteq W_2$ reads “ W_2 is coarser”.

Our notion is deliberately weaker than the standard homotopical-category one: the weak equivalences of a homotopical category are required to satisfy two-out-of-six and, in particular, to contain all isomorphisms (Dwyer–Hirschhorn–Kan–Smith [5]). We impose only two-out-of-three and do *not* require $\text{Iso}(C) \subseteq W$, so that $\{\text{id}\} < \text{Iso}(C)$ is allowed; this is what makes $\text{Same}(C)$ a fine invariant — see the localization discussion below and the group case (§3).

The identities form the least sameness $\perp = \{\text{id}\}$; the isomorphisms $\text{Iso}(C)$ form a distinguished sameness, in general strictly above $\perp$ (axiom (iii) makes $\text{Iso}(C)$ a sameness, since two invertible among $f, g, g \circ f$ force the third invertible). Each W determines a localization $C \rightarrow C[W^{-1}]$, but distinct samenesses may present equivalent localizations (both $\{\text{id}\}$ and $\text{Iso}(C)$ localize to C); thus $\text{Same}(C)$ is the lattice of *weak-equivalence structures*, which surjects onto, but is finer than, the poset of localizations. It is this finer, purely arrow-theoretic object that carries the structure below.

Contributions.

- $\text{Same}(C)$ is a complete lattice (§2).
- $\text{Same}(G) = \text{Sub}(G)$ for a group G (§3); hence M_3 (Klein four) and non-distributive, non-modular sameness lattices occur, and the realization problem for Same contains that for subgroup lattices.
- structure and reconstruction (§4): a product decomposition over connected components, a reconstruction obstruction within a component, and the lattice position (non-modular in general; Ore on groups).
- $\text{Same}: \mathbf{Cat} \rightarrow \mathbf{CLat}_{\text{Gal}}$ is a functor into Galois connections (§5): $F \mapsto (F_! \dashv F^*)$ with $(GF)^* = F^*G^*$ and $(GF)_! = G_!F_!$.

- comparison intervals, the localization factorization at $V_F = F^*(\text{Iso}D)$, and a conservativity theorem for it under a calculus of fractions (§6).
- the cubical-site phase diagram as a map into $\text{Same}(\text{Presh}(S))$ (§7), and an object-oriented reading with the Structure Identity Principle as one level (§8).

Every finite combinatorial claim is verified by exhaustive computation; the scripts are named inline and listed in §9.

Related work. Weak equivalences are classically studied *together with additional homotopical structure*. A category of fractions and its calculus invert a chosen class (Gabriel–Zisman [2]); a relative category is a category with a distinguished weak-equivalence subcategory (Barwick–Kan [4]); a homotopical category adds two-out-of-six and all isomorphisms (Dwyer–Hirschhorn–Kan–Smith [5]); Cisinski’s localizers on a presheaf topos form their own hierarchy, closed under intersection with a smallest generated localizer [6]; and Balchin, Ormsby, Osorno and Roitzheim [7] *enumerate all model structures* on a finite total order $[n]$, a rich collection organised by the Catalan triangle. So collections of weak-equivalence data are already studied. What is new here is the object of study: we isolate the *bare* identity/composition/ two-out-of-three classes themselves — independently of cofibrations, fibrations, factorisation systems, or saturation — and organise *all* of them into a single functorial complete lattice $\text{Same}: \mathbf{Cat} \rightarrow \mathbf{CLat}_{\text{Gal}}$. The subgroup-lattice identification on groups (Theorem 3.1) and groupoids (Theorem 3.3), the Galois connection a functor induces (Theorem 5.2), and the localization/conservativity results (§6) are facts about this lattice; the cubical-site comparison (§7) is an instance within it. Subgroup lattices and their representation are themselves a central theme of lattice theory (Ore [3]; Pálffy–Pudlák [8]).

2. THE SAMENESS LATTICE

Proposition 2.1. *$\text{Same}(C)$ is a complete lattice. Meets are intersections, the top is $\text{Mor}(C)$, and the join of a family is the sameness generated by its union.*

Proof. Axioms (i)–(iii) are each preserved by arbitrary intersection: if two of $\{f, g, g \circ f\}$ lie in every W_i then the third lies in each, hence in the intersection; identities and composition-closure are clear. $\text{Mor}(C)$ satisfies (i)–(iii). A meet-complete poset with a top is a complete lattice; joins are the meets of upper bounds. $\square$

Example 2.2 (the free composition triangle). Let C be the poset chain $0 < 1 < 2$: the non-identity maps are $f: 0 \rightarrow 1$, $g: 1 \rightarrow 2$ and $h = g \circ f$, with no other relations and no non-identity isomorphisms. Two-out-of-three forbids a sameness of exact weight two among $\{f, g, h\}$, so, relative

to the three non-identity morphisms, the possibilities are $\emptyset$, the three singletons $\{f\}, \{g\}, \{h\}$, and $\{f, g, h\}$ — each taken together with the identities: $\text{Same}(C) \cong M_3$, the five-element non-distributive modular lattice (Figure 1). Morphisms sharing no composite instead impose no constraint and contribute independent Boolean factors; e.g. the parallel pair $a \rightrightarrows b$ and the span $c \leftarrow a \rightarrow b$ each give the Boolean lattice 2^2 . (`sameness_lattice.py`: $|\text{Same}(\text{3-chain})| = 5$, $|\text{Same}(\text{4-chain})| = 15$.)

Remark 2.3. The M_3 of Example 2.2 arises from the *free* triangle. It is not true that every composition triangle produces an M_3 : on the cyclic group C_4 (a one-object category with many composites) the sameness lattice is a chain (Example 3.2). Thus $\text{Same}(C)$ detects certain *independent* composition configurations rather than all composition; the group case makes the correct statement precise.

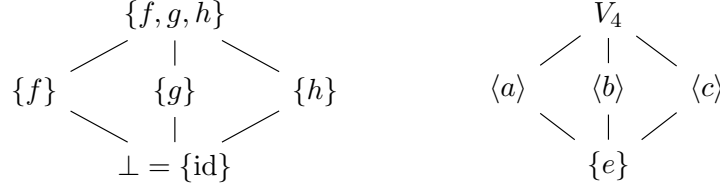


FIGURE 1. The same lattice M_3 from two sources: the free composition triangle $0 < 1 < 2$ (left, Example 2.2) and the Klein four-group V_4 (right, $\text{Same}(V_4) = \text{Sub}(V_4)$, Theorem 3.1).

3. THE SUBGROUP-LATTICE THEOREM

Theorem 3.1. *Let G be a group, viewed as a one-object category. Then $\text{Same}(G) = \text{Sub}(G)$, the lattice of subgroups of G .*

Proof. A subgroup $H \leq G$ contains e , is closed under multiplication, and satisfies two-out-of-three (if two of f, g, gf lie in H the third is a product of them and their inverses, hence in H); so H is a sameness. Conversely let W be a sameness. It contains e and is closed under multiplication. It is closed under inverses: for $w \in W$, the composable pair (w^{-1}, w) has composite $w^{-1}w = e \in W$ and factor $w \in W$, so two of $\{w, w^{-1}, e\}$ lie in W , whence two-out-of-three gives $w^{-1} \in W$. A subset of a group containing e and closed under multiplication and inverses is a subgroup. Hence $W \in \text{Sub}(G)$. The orders coincide with inclusion, so $\text{Same}(G) = \text{Sub}(G)$ as lattices. $\square$

Example 3.2. $\text{Same}(C_4) = \text{Sub}(C_4)$ is the chain $\{e\} < \{e, a^2\} < C_4$, hence distributive, although C_4 has composition triangles (Remark 2.3). $\text{Same}(V_4) = \text{Sub}(V_4) \cong M_3$ (Figure 1, right), non-distributive. $\text{Same}(S_3) = \text{Sub}(S_3)$ has four atoms (three of order 2, one of order 3). All verified in `same_of_group.py`.

Theorem 3.1 is not an accident of the one-object case; it is the one-object instance of a groupoid statement.

Theorem 3.3. *Let $\mathcal{G}$ be a groupoid. Then $\text{Same}(\mathcal{G})$ is the lattice of wide subgroupoids of $\mathcal{G}$ (subgroupoids containing every object), ordered by inclusion.*

Proof. Let W be a sameness. As in Theorem 3.1, two-out-of-three forces inverse-closure: for $w \in W$ the pair (w^{-1}, w) has composite an identity in W and factor $w \in W$, so $w^{-1} \in W$. With (i) all identities, (ii) composition-closure and inverse-closure, W is a subgroupoid; and it is wide because it contains id_x for every object x . Conversely a wide subgroupoid contains the identities, is closed under composition, and satisfies two-out-of-three (being closed under composition and inverses), hence is a sameness. Order by inclusion agrees. Verified on the walking-isomorphism groupoid in `same_conservative_cx.py`. $\square$

Corollary 3.4. *Every subgroup lattice of a group occurs as a sameness lattice (Theorem 3.1), so the class $\{\text{Same}(C)\}$ is not contained in the distributive (nor the modular) lattices, and the problem of characterising which lattices arise as $\text{Same}(C)$ contains the subgroup-lattice representation problem.*

Pálffy and Pudlák [8] proved that the finite lattice representation problem (whether every finite lattice is the congruence lattice of a finite algebra) is equivalent to the question — still open — of whether every finite lattice occurs as an interval in the subgroup lattice of a finite group. Via Theorem 3.1 every such interval is an interval of a sameness lattice, so the sameness lattices form a class at least as rich as the subgroup lattices — a first structural reason to regard Same as a genuine invariant rather than a formal device.

4. STRUCTURE AND RECONSTRUCTION

We record what $\text{Same}(C)$ does and does not remember of C , and where its lattice sits.

Theorem 4.1 (component decomposition). *If $C = \coprod_i C_i$ is the decomposition into connected components, then*

$$\text{Same}(C) \cong \prod_i \text{Same}(C_i)$$

as complete lattices.

Proof. A morphism of C lies in a unique component, and composites and the pairs witnessing two-out-of-three stay within a component (there are no morphisms between components). Hence $W \subseteq \text{Mor}(C)$ is a sameness iff each $W \cap \text{Mor}(C_i)$ is a sameness of C_i , and $W \mapsto (W \cap \text{Mor}(C_i))_i$ is an order isomorphism onto $\prod_i \text{Same}(C_i)$. Verified on $C_1 \amalg C_2$ with C_1 the 3-chain and C_2 the 2-chain: $|\text{Same}(C_1 \amalg C_2)| = 10 = 5 \cdot 2$ (`same_structure.py`). $\square$

So Same turns the coproduct of categories into the product of lattices. It does *not* thereby recover the components: a connected category can already have a decomposable sameness lattice (the parallel pair and the span give $\text{Same} \cong 2^2 \cong 2 \times 2$), so the product factorization of $\text{Same}(C)$ is finer than, and does not determine, the component decomposition of C . Within a single component the invariant is in any case genuinely lossy.

Theorem 4.2 (reconstruction obstruction). *Same does not determine C , even up to the crudest features. The free composition triangle (the 3-object poset $0 < 1 < 2$) and the Klein four-group V_4 (a one-object group) satisfy*

$$\text{Same}(0 < 1 < 2) \cong M_3 \cong \text{Same}(V_4),$$

although the two categories differ in their number of objects, in whether every morphism is invertible, and in being a poset versus a group.

Proof. Example 2.2 and Theorem 3.1 with $\text{Sub}(V_4) \cong M_3$; both computed in `sameness_lattice.py`, `same_of_group.py`. $\square$

Thus a reconstruction theorem for Same must be relative to extra data (a connected, or connected-and-rigid, hypothesis); the raw invariant is a genuine lattice-valued measurement, not an encoding of C . Its lattice-theoretic position is correspondingly broad.

Theorem 4.3 (lattice position). *The class $\{\text{Same}(C)\}$ is not contained in the modular lattices, hence not in the distributive ones. Concretely $\text{Same}(V_4) \cong M_3$ is modular but not distributive, and $\text{Same}(A_4) = \text{Sub}(A_4)$ is not modular (it contains a pentagon N_5). On a group, by Ore’s theorem, $\text{Same}(G) = \text{Sub}(G)$ is distributive iff G is locally cyclic (for finite G , iff G is cyclic).*

Proof. $\text{Same}(V_4) \cong M_3$ is modular non-distributive. For A_4 : with $a = \langle(12)(34)\rangle$ (order 2), $b = V_4 \leq A_4$ (order 4, $a \leq b$) and $x = \langle(123)\rangle$ (order 3), one has $x \wedge b = \{e\}$ and $a \vee x = A_4$, so $a \vee (x \wedge b) = a \neq b = (a \vee x) \wedge b$: the modular law fails, an N_5 . The equality $\text{Same}(A_4) = \text{Sub}(A_4)$, and the failure of modularity, are verified exhaustively in `same_structure.py`. The distributive characterisation is Ore’s theorem [3] applied through Theorem 3.1; it is consistent with $\text{Same}(C_4)$ a chain (cyclic) and $\text{Same}(V_4), \text{Same}(S_3)$ non-distributive (non-cyclic), all computed in Example 3.2. $\square$

Remark 4.4. Theorems 4.1–4.3 answer, in part, the two questions the invariant raises. Reconstruction: $\text{Same}(C)$ is *compatible* with the connected-component decomposition (it sends it to a lattice product, Theorem 4.1) but recovers neither the components nor anything finer in general (Theorem 4.2, and the decomposable Same of a connected category). Classification: distributivity of Same is pinned down on groups by Ore (Theorem 4.3), while a full category-level characterisation of distributive or modular $\text{Same}(C)$ remains open and, by Corollary 3.4, at least as hard as the corresponding subgroup-lattice questions.

5. GALOIS FUNCTORIALITY

Proposition 5.1. *For a functor $F: C \rightarrow D$ and $W \in \text{Same}(D)$ set $F^*(W) = \{m \in \text{Mor}(C) : F(m) \in W\}$. Then $F^*(W) \in \text{Same}(C)$, and $F^*: \text{Same}(D) \rightarrow \text{Same}(C)$ preserves arbitrary meets; hence it has a left adjoint $F_! (V) = \bigwedge \{W : V \subseteq F^*(W)\}$, giving a Galois connection $F_! \dashv F^*$ of complete lattices, $V \subseteq F^*(W) \Leftrightarrow F_!(V) \subseteq W$.*

Proof. $F(1) = 1$; $F(g \circ f) = F(g) \circ F(f)$; and if two of $f, g, g \circ f$ lie in $F^*(W)$ then two of their images lie in W , so the third does, so the remaining map lies in $F^*(W)$: thus $F^*(W)$ is a sameness. $F^*(\bigcap_i W_i) = \bigcap_i F^*(W_i)$, so F^* preserves meets; a meet-preserving map of complete lattices has the displayed left adjoint. $\square$

Let $\mathbf{CLat}_{\text{Gal}}$ be the category whose objects are complete lattices and whose morphisms $L \rightarrow M$ are Galois connections, i.e. monotone adjunctions $\ell \dashv r$ with $\ell: L \rightarrow M$ left adjoint to $r: M \rightarrow L$; composition is composition of adjunctions, $(\ell' \dashv r') \circ (\ell \dashv r) = (\ell' \ell \dashv r r')$, and the identity is $(\text{id} \dashv \text{id})$. A morphism is determined by either adjoint, so $\mathbf{CLat}_{\text{Gal}}$ is isomorphic both to the category of complete lattices with left (join-preserving) maps and to its opposite via right (meet-preserving) maps.

Theorem 5.2. *Same: $\mathbf{Cat} \rightarrow \mathbf{CLat}_{\text{Gal}}$ is a functor: $C \mapsto \text{Same}(C)$ on objects, and a functor $F: C \rightarrow D$ to the Galois connection $F_! \dashv F^*$. In particular, for $C \xrightarrow{F} D \xrightarrow{G} E$,*

$$(GF)^* = F^*G^*, \quad (GF)_! = G_!F_!, \quad (\text{id}_C)^* = \text{id} = (\text{id}_C)_!.$$

Proof. $F_! \dashv F^*$ is a morphism of $\mathbf{CLat}_{\text{Gal}}$ (Proposition 5.1). $(GF)^*(W) = \{m : GF(m) \in W\} = \{m : F(m) \in G^*(W)\} = F^*(G^*(W))$, and $(\text{id})^* = \text{id}$; so $F \mapsto F^*$ is functorial on right adjoints. Left adjoints are unique and compose to the composite's left adjoint, so $(GF)_! = G_!F_!$ and $(\text{id})_! = \text{id}$; this is exactly functoriality into $\mathbf{CLat}_{\text{Gal}}$. Verified exhaustively on $(0 < 1 < 2) \rightarrow (0 < 1) \rightarrow *$ in `same_functoriality.py`. $\square$

Thus “a functor induces a Galois connection” is not an isolated observation but the action of a functor Same (Figure 2): F^* pulls a viewpoint on D back to C , $F_!$ pushes forward, and the two compose along $\mathbf{Cat}$ as displayed.

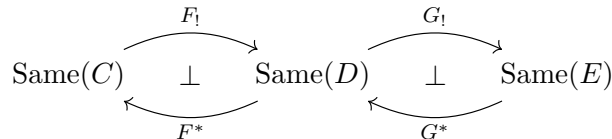


FIGURE 2. Same: $\mathbf{Cat} \rightarrow \mathbf{CLat}_{\text{Gal}}$ (Theorem 5.2); composites satisfy $(GF)^* = F^*G^*$, $(GF)_! = G_!F_!$.

6. COMPARISON INTERVALS AND THE LOCALIZATION FACTORIZATION

Theorem 6.1 (pair comparison). *For $W_1, W_2 \in \text{Same}(C)$ the comparison interval is $[W_1 \wedge W_2, W_1 \vee W_2] \subseteq \text{Same}(C)$, with $W_1 \wedge W_2$ the finest common sameness and $W_1 \vee W_2$ the coarsest common coarsening; the two agree, $W_1 = W_2$, iff the interval is a single point. If $W_1 \subseteq W_2$ the interval is $[W_1, W_2]$.*

Proof. Meet and join exist (Proposition 2.1) and bound both points; $W_1 = W_2$ iff meet = join. $\square$

Proposition 6.2 (largest inverted sameness). *Let $F: C \rightarrow D$ and put $V_F := F^*(\text{Iso}(D)) = \{m \in \text{Mor}(C) : F(m) \text{ is an isomorphism}\} \in \text{Same}(C)$. Then V_F is the largest sameness F sends into isomorphisms: $F(V) \subseteq \text{Iso}(D) \iff V \subseteq V_F$. Consequently F factors, uniquely up to isomorphism, through the localization, $C \xrightarrow{L_{V_F}} C[V_F^{-1}] \xrightarrow{\bar{F}} D$. The interval $[\text{Iso}(C), V_F] \subseteq \text{Same}(C)$ measures the collapse of F on the morphisms of C , and is trivial iff F reflects isomorphisms among morphisms of C .*

Proof. $V_F = F^*(\text{Iso}D)$ is a sameness (Proposition 5.1); F sends a sameness V into $\text{Iso}(D)$ iff $V \subseteq V_F$, so V_F is the largest such. F inverts V_F , so it factors through L_{V_F} by the universal property of localization. $\text{Iso}(C) \subseteq V_F$ always, with equality iff $F(m)$ iso implies m iso for $m \in \text{Mor}(C)$. Verified ($V_F = \{g\}$ for $(0 < 1 < 2) \rightarrow (0 < 1)$) in `comparison_factorization.py`. $\square$

The factorising functor $\bar{F}$ is, however, *not* conservative in general: V_F controls only the morphisms of C , whereas localization adds new zigzag morphisms that F may send to isomorphisms.

Proposition 6.3 (the factorization need not be conservative). *There is a functor $F: C \rightarrow D$ for which $\bar{F}: C[V_F^{-1}] \rightarrow D$ does not reflect isomorphisms.*

Proof. Let C be freely generated by $g: Z \rightarrow X$, $f: X \rightarrow Y$ and $v: Z \rightarrow Y$, and let D present a retract, $i: A \rightarrow B$, $r: B \rightarrow A$ with $ri = 1_A$ and $ir \neq 1_B$. Put $F(X) = A$, $F(Y) = F(Z) = B$, $F(f) = i$, $F(g) = r$, $F(v) = 1_B$. Then $F(v)$ is an isomorphism while $F(f), F(g)$ are not, so $V_F = \langle v \rangle$ and $f, g \notin V_F$. In $C[V_F^{-1}]$ the morphism $\alpha = g \circ v^{-1} \circ f: X \rightarrow X$ has $\bar{F}(\alpha) = r \circ 1_B^{-1} \circ i = ri = 1_A$, an isomorphism; but α is not invertible in $C[V_F^{-1}]$ — the functor to **Set** sending v to a bijection and f to a constant map inverts v yet makes α a constant, non-bijective map. So $\bar{F}$ sends the non-isomorphism α to an isomorphism. All data are checked in `same_conservative_cx.py`. $\square$

Conservativity of $\bar{F}$ is thus a genuine extra condition. The failure in Proposition 6.3 comes from a length-three zigzag that cannot be straightened into a single fraction; ruling this out restores conservativity.

Theorem 6.4 (conservativity under a calculus of fractions). *If V_F admits a calculus of left fractions (or, dually, of right fractions) in C , then $\bar{F}: C[V_F^{-1}] \rightarrow D$ is conservative.*

Proof. Under a left calculus of fractions every morphism of $C[V_F^{-1}]$ is $s^{-1} \circ a$ with $a \in \text{Mor}(C)$ and $s \in V_F$ [2]. Suppose $\bar{F}(s^{-1}a) = F(s)^{-1}F(a)$ is an isomorphism of D . Since $s \in V_F$, $F(s)$ is an isomorphism, so $F(a)$ is an isomorphism, i.e. $a \in V_F$. But every morphism of V_F is inverted in $C[V_F^{-1}]$, so both a and s become isomorphisms there, hence $s^{-1}a$ is an isomorphism of $C[V_F^{-1}]$. Thus $\bar{F}$ reflects isomorphisms. The dual argument covers right fractions. $\square$

The hypothesis is exactly what the counterexample lacks: for $C, V_F = \langle v \rangle$ of Proposition 6.3 the square-completion axiom fails already for $s = v, a = g$ (there is no morphism $Y \rightarrow X$ to complete the square), so V_F admits no calculus of left fractions (`same_conservative.cx.py`) — consistent with $\bar{F}$ being non-conservative there. The extreme case is when the factorization is not merely conservative but an equivalence:

Proposition 6.5 (when the factorization is an equivalence). $\bar{F}: C[V_F^{-1}] \rightarrow D$ is an equivalence iff F exhibits D as the localization of C at V_F (i.e. F is initial among functors inverting V_F). In that case $\bar{F}$ is in particular conservative.

Proof. $C \rightarrow C[V_F^{-1}]$ is initial among functors inverting V_F ; F inverts V_F and factors as $\bar{F} \circ L_{V_F}$. If F is also initial with this property, the two initial factorizations are equivalent, so $\bar{F}$ is an equivalence; conversely an equivalence $\bar{F}$ makes $F = \bar{F} \circ L_{V_F}$ initial. An equivalence reflects isomorphisms. $\square$

Remark 6.6 (two open problems). Theorem 6.4 gives a sufficient condition for $\bar{F}$ to be conservative; a full characterisation of the F for which $\bar{F}$ is conservative remains open, and is in our view the central refinement this factorization invites. A second, strictly stronger problem is to characterise the levels $W \in \text{Same}(D)$ at which F is a W -equivalence, meaning $\bar{F}_W: C[F^*W^{-1}] \rightarrow D[W^{-1}]$ is an equivalence; this is not the same question, since an equivalence is conservative but conservativity is far from an equivalence — it does not even imply faithfulness (the functor collapsing a parallel pair $X \rightrightarrows Y$ to one arrow is full, essentially surjective and conservative but not faithful, hence not an equivalence, `same_functoriality.py`).

Finally, the pair comparison of Theorem 6.1 is *not* the same as comparing localizations, and this is a feature. A pair $W_1 \subseteq W_2$ spans a possibly non-trivial interval $[W_1, W_2]$ even when $C[W_1^{-1}]$ and $C[W_2^{-1}]$ agree: already $W_1 = \{\text{id}\}$ and $W_2 = \text{Iso}(C)$ are distinct samenesses with equivalent localizations. Hence $\text{Same}(C)$ is strictly finer than the poset of localizations, and *pair comparison records more than localization comparison* — the extra information being exactly which morphisms, not merely which localization, one has chosen to invert.

7. THE CUBICAL-SITE PHASE DIAGRAM AS A MAP INTO Same

Definition 1.1 was stated for a small category, but $\text{Presh}(S)$ is not small. We fix Grothendieck universes $U \in V$, take S and its presheaves to be

U -small, and read $\text{Same}(\text{Presh}(S))$ as the (V -small) lattice of U -definable samenesses; the two model-structure classes below are of this form, so the discussion is size-safe.

This section is an *illustration*, not a source of new theorems: it reads results established in the companion manuscript [1] through the lattice of the present paper. Let S be a cube site and $C = \text{Presh}(S)$; the cubical-type and Cisinski test model structures give two samenesses $W_{\text{type}} \subseteq W_{\text{test}}$ on every standard site [1]. By Theorem 6.1 they span an interval, and the classification “which cube sites present spaces” becomes a single map

$$S \longmapsto [W_{\text{type}}, W_{\text{test}}] \in \text{Same}(\text{Presh}(S)),$$

under which S *presents spaces* iff $W_{\text{type}} = W_{\text{test}}$, i.e. iff the interval is trivial. The gap $W_{\text{test}} \setminus W_{\text{type}}$ is the set of isotropy witnesses. On the three reversal sites it is stratified by a reflection-subgroup criterion [1]: for a mixed $H \leq B_n$ (hyperoctahedral group) with $N = \langle\langle \text{Ref}(H) \rangle\rangle$ the normal closure of its coordinate reversals, a test-not-type weak equivalence is attached to H iff N is mixed. Running this criterion (`reflection_witness.py`) gives, among subgroups generated by at most two elements, 4 witness-bearing $H \leq B_2$ and $26 \leq B_3$; pure-permutation isotropy never separates the two levels. Table 1 records the settled entries; two routes to a trivial interval are distinguished by the lattice picture: on the one-connection site the gap is *emptied by an equivariant contraction*, while on the Dedekind site the gap has *no quotient generators* (the monotone dunce hat is contractible in both structures), leaving a regularity question — the one open entry.

cube site S	interval $[W_{\text{type}}, W_{\text{test}}]$	presents spaces?
cartesian	non-trivial	No
one connection	trivial (gap emptied)	Yes
Dedekind	open (conj. trivial)	open
De Morgan	non-trivial	No
Kleene	non-trivial	No
Boolean	non-trivial	No

TABLE 1. The settled part of the phase diagram as values of $S \mapsto [W_{\text{type}}, W_{\text{test}}]$; “presents spaces” is a trivial interval. The Dedekind entry is open.

8. AN OBJECT-ORIENTED READING

The lattice supports a reading in which an object is known only up to a chosen level, in the spirit of the Structure Identity Principle and of coalgebraic behavioural equivalence. Interpret a sameness W by the observation quotient

$$\llbracket C \rrbracket_W := \text{Ob}(C) / \sim_W,$$

where $a \sim_W b$ iff a, b are linked by a zigzag of morphisms in W , with $\pi_W: \text{Ob}(C) \twoheadrightarrow \llbracket C \rrbracket_W$. A W -observable is a map on objects that factors through π_W (is constant on $\sim_W$ -classes): observability at level W is factorization through the interface $\llbracket C \rrbracket_W$ (Figure 3).

Proposition 8.1. *$W \subseteq W'$ makes $\llbracket C \rrbracket_{W'}$ a further quotient of $\llbracket C \rrbracket_W$, so every W' -observable is a W -observable (the interface narrows as the level coarsens); the assignment $W \mapsto \pi_W$ is a map of posets from $\text{Same}(C)$ to quotients of $\text{Ob}(C)$. At the sameness $W = \text{Iso}(C)$ the relation $\sim_W$ is “isomorphic”, and the $\text{Iso}(C)$ -observables are the isomorphism-invariants: this level is the Structure Identity Principle. For $W_{\text{type}} \subseteq W_{\text{test}}$ on $\text{Presh}(S)$, “ a, b present the same space” is a $\sim_{W_{\text{test}}}$ b , and if S presents spaces ($W_{\text{type}} = W_{\text{test}}$) then $\llbracket C \rrbracket_{W_{\text{type}}} = \llbracket C \rrbracket_{W_{\text{test}}}$; the converse is not automatic, as distinct samenesses can induce the same object-level equivalence, and needs the Hom-level refinement of Remark 8.2.*

Proof. A zigzag in W is a zigzag in $W' \supseteq W$, so $\sim_W$ refines $\sim_{W'}$ and $\pi_{W'}$ factors through π_W ; a map through $\pi_{W'}$ is a map through π_W . The remaining statements are the definitions, with the cube-site clause from §7. Verified on the 3-chain in `objectoriented.py` ($\perp \mapsto \{0\}\{1\}\{2\}$ up to $\top \mapsto \{012\}$, monotone). $\square$

$$\begin{array}{ccc} \text{Ob}(C) & \xrightarrow{\pi_W} & \llbracket C \rrbracket_W \\ \pi_{W'} \downarrow & \swarrow W \leq W' & \\ \llbracket C \rrbracket_{W'} & & \end{array}$$

FIGURE 3. Encapsulation and level shift (Proposition 8.1): a W -observable factors through the interface π_W ; coarsening quotients further. The level $W = \text{Iso}(C)$ is the Structure Identity Principle.

Remark 8.2. This is a semantic construction, not yet a proof theory. Upgrading $\sim_W$ from object zigzags to the full localization $C[W^{-1}]$ (Hom-level behavioural equivalence, matching the coalgebraic bisimulation reading) refines the interface and, in particular, is what a converse to the object-level “presents spaces” implication of Proposition 8.1 requires. A graded sequent calculus over $(\text{Same}(C), \subseteq)$ with an S4 level-shift modality, and its completeness and cut-elimination, are developed elsewhere.

9. COMPUTATIONAL RECORD

All finite claims are verified by exhaustive computation in `scripts/comparison/`: `sameness_lattice.py` (Proposition 2.1, Example 2.2), `same_of_group.py` (Theorem 3.1, Example 3.2),

`same_structure.py` (Theorems 4.1–4.3), `same_functoriality.py` (Theorem 5.2 and Remark 6.6), `comparison_factorization.py`, `same_conservative_cx.py` (Theorem 3.3, Proposition 6.3, and the calculus-of-fractions failure behind Theorem 6.4) and `interval_witness.py` (Theorem 6.1, Proposition 6.2), `reflection_witness.py` (§7), and `objectoriented.py` (Proposition 8.1).

10. CONCLUSION

Same is a functorial invariant of a category: a complete lattice, the subgroup lattice on groups (and the wide-subgroupoid lattice on groupoids), and a functor into Galois connections, within which a comparison of theories is an interval and a functor factors through the localization at its largest inverted sameness. Section 4 settles the first layer of two natural problems. The *reconstruction* problem — how much of C is determined by $\text{Same}(C)$ — is compatible with the component decomposition (which it sends to a lattice product, Theorem 4.1) but does not recover the components, and is genuinely lossy within a component (Theorem 4.2); a sharp reconstruction under a rigidity hypothesis remains to be found. The *classification* problem — which C have $\text{Same}(C)$ distributive, modular, or semidistributive — is pinned down on groups by Ore (Theorem 4.3) but open at the level of categories, and by Corollary 3.4 at least as hard as the subgroup-lattice representation problem. Both are squarely lattice-theoretic questions raised by a categorical construction, and either would deepen the invariant further.

DECLARATIONS

Funding. No funding was received for this work.

Competing interests. The author declares that he has no competing interests.

Corresponding author. Ryota Shibusawa, Daiichi Institute of Technology; e-mail: `r-shibusawa@daiichi-koudai.ac.jp`.

Availability of data and materials. The computational verification scripts named in §9 are available from the author on request.

REFERENCES

- [1] R. Shibusawa, *Which cubical sites present spaces? Reflection subgroups, connections, and the comparison of the type and test model structures*, companion manuscript, under review (2026).
- [2] P. Gabriel, M. Zisman, *Calculus of Fractions and Homotopy Theory*, Ergeb. Math. Grenzgeb. **35**, Springer, 1967.
- [3] O. Ore, *Structures and group theory. II*, Duke Math. J. **4** (1938), 247–269.
- [4] C. Barwick, D. M. Kan, *Relative categories: another model for the homotopy theory of homotopy theories*, Indag. Math. **23** (2012), 42–68.
- [5] W. G. Dwyer, P. S. Hirschhorn, D. M. Kan, J. H. Smith, *Homotopy Limit Functors on Model Categories and Homotopical Categories*, Math. Surveys Monogr. **113**, Amer. Math. Soc., 2004.

- [6] D.-C. Cisinski, *Les préfaisceaux comme modèles des types d'homotopie*, Astérisque **308**, Soc. Math. France, 2006.
- [7] S. Balchin, K. Ormsby, A. M. Osorno, C. Roitzheim, *Model structures on finite total orders*, Math. Z. **304** (2023), article no. 40.
- [8] P. P. Pálffy, P. Pudlák, *Congruence lattices of finite algebras and intervals in subgroup lattices of finite groups*, Algebra Universalis **11** (1980), 22–27.
- [9] J. J. M. M. Rutten, *Universal coalgebra: a theory of systems*, Theoret. Comput. Sci. **249** (2000), 3–80.
- [10] The Univalent Foundations Program, *Homotopy Type Theory: Univalent Foundations of Mathematics*, IAS, 2013.

DAIICHI INSTITUTE OF TECHNOLOGY

Email address: `r-shibusawa@daiichi-koudai.ac.jp`